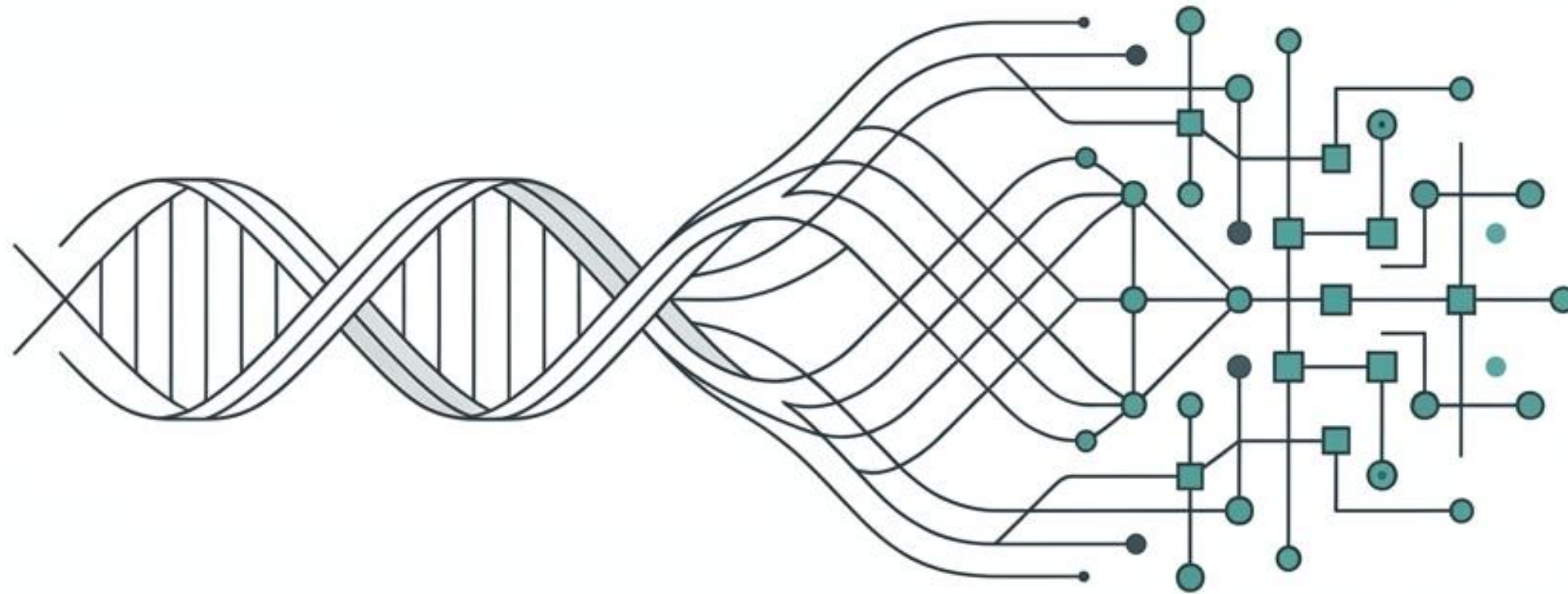
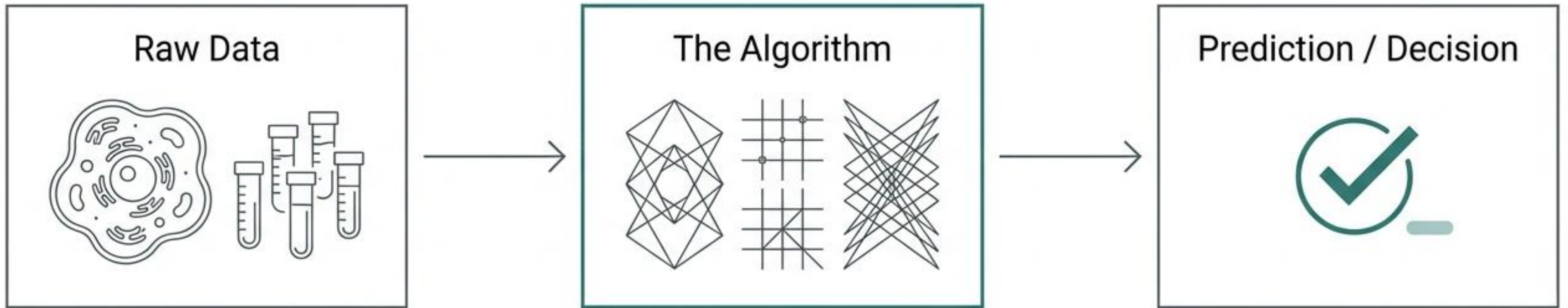




Machine Learning in Bioinformatics

Assoc.Prof. Dr. Wararat Songpan, College of Computing, Khon Kaen University Thailand

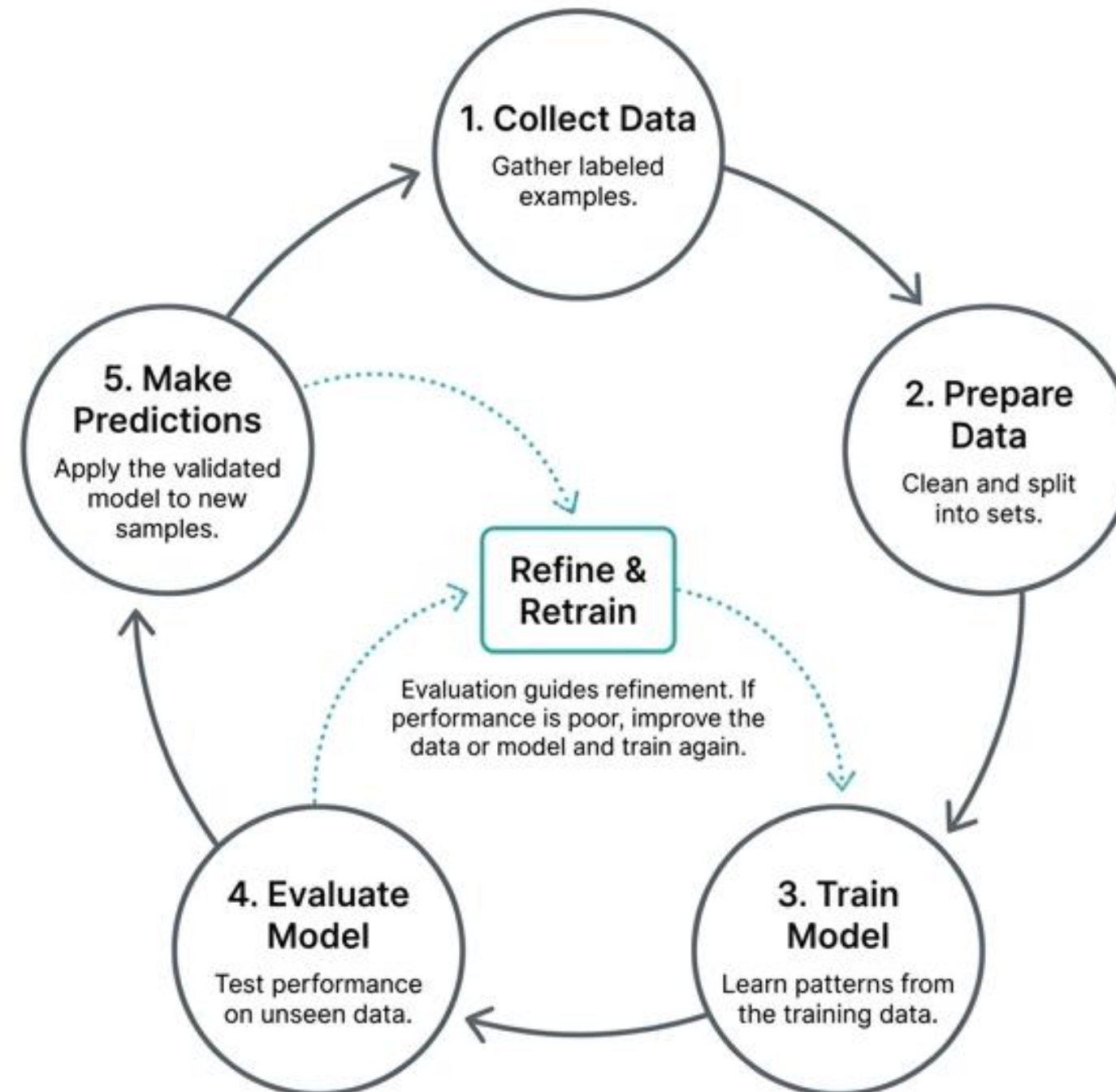
Discovering the Rules



Machine learning uses algorithms that learn patterns from data to make predictions on unseen samples.

We do not explicitly program the diagnostic rules; we provide the biological measurements, and the model discovers the underlying patterns to interpret computational evidence responsibly.

The Machine Learning Lifecycle



Mapping the Algorithm Landscape

	Supervised Learning	Unsupervised Learning
Mechanism	Learns from labeled data.	Finds hidden structure in data without known labels.
Bioinformatics Application	• Classification: Predicting disease vs. non-disease from patient data. • Regression: Predicting continuous values like drug response.	• Clustering: Discovering novel tumor subtypes. • Dimensionality Reduction: Summarizing high-dimensional omics data.

Translating Biology into Variables



Features

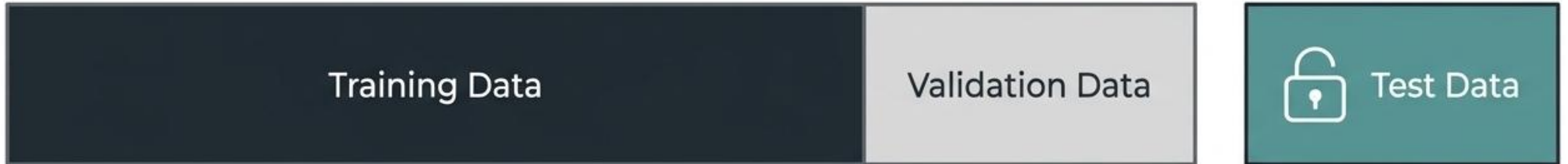
The biological sample.
Examples: Gene expression,
protein levels, clinical
measurements.

Target

The outcome.
Examples: Disease class,
phenotype, drug response.

In your specific research problem, what exactly are your X and y?

Protecting the Future: The Golden Rule of Data Splitting



Training Data:

Used to learn and fit the model parameters.

Validation / Cross-Validation:

Used to compare choices, tune parameters, and select the best model.

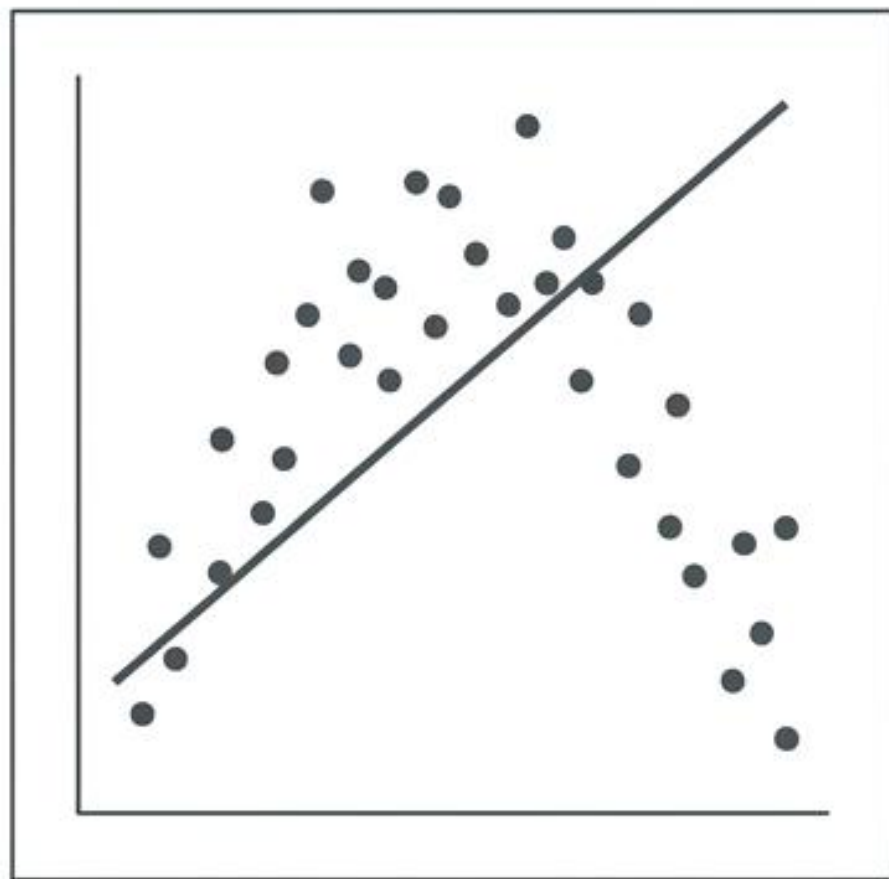
Test Data:

The **ultimate vault**. Used solely for final, unbiased evaluation on unseen data.

CRUCIAL WARNING:

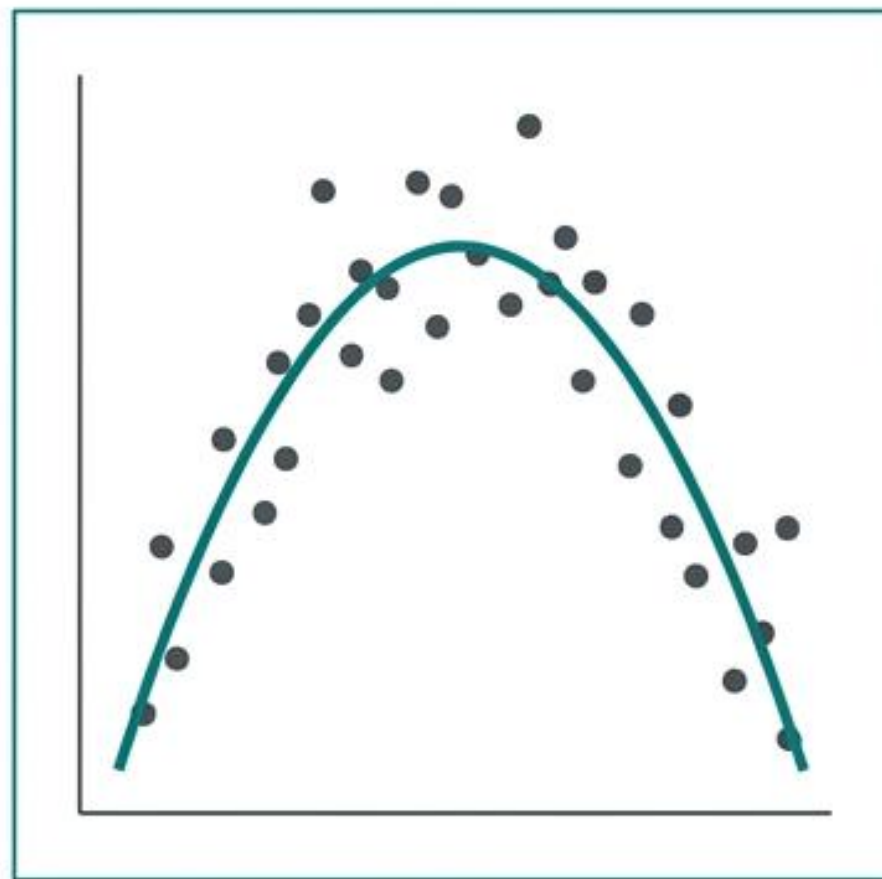
The **test set must NEVER** influence training, model selection, or preprocessing decisions. Doing so results in dangerously optimistic performance reports.

The Goldilocks Principle of Learning



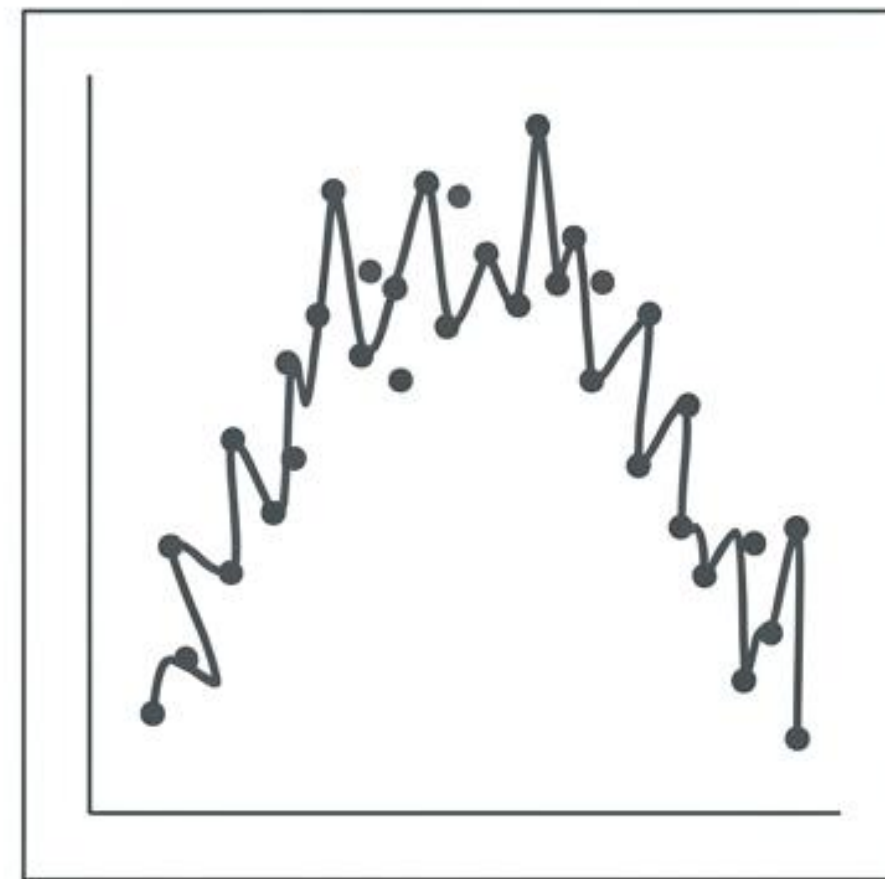
Underfitting

The model is too simple to capture meaningful biological patterns. Fails even on training data.



Good Generalization

Learns useful patterns and performs consistently well on unseen samples from the broader population.



Overfitting

Memorizes training-specific noise. Displays excellent training performance but collapses when exposed to new patients.

Beyond Basic Accuracy: The Confusion Matrix

Context: 100 patient samples. Positive = disease present; Negative = disease absent.

		Actual Result	
Predicted Result	Positive	36 True Positive	6 False Positive
	Negative	4 False Negative	54 True Negative

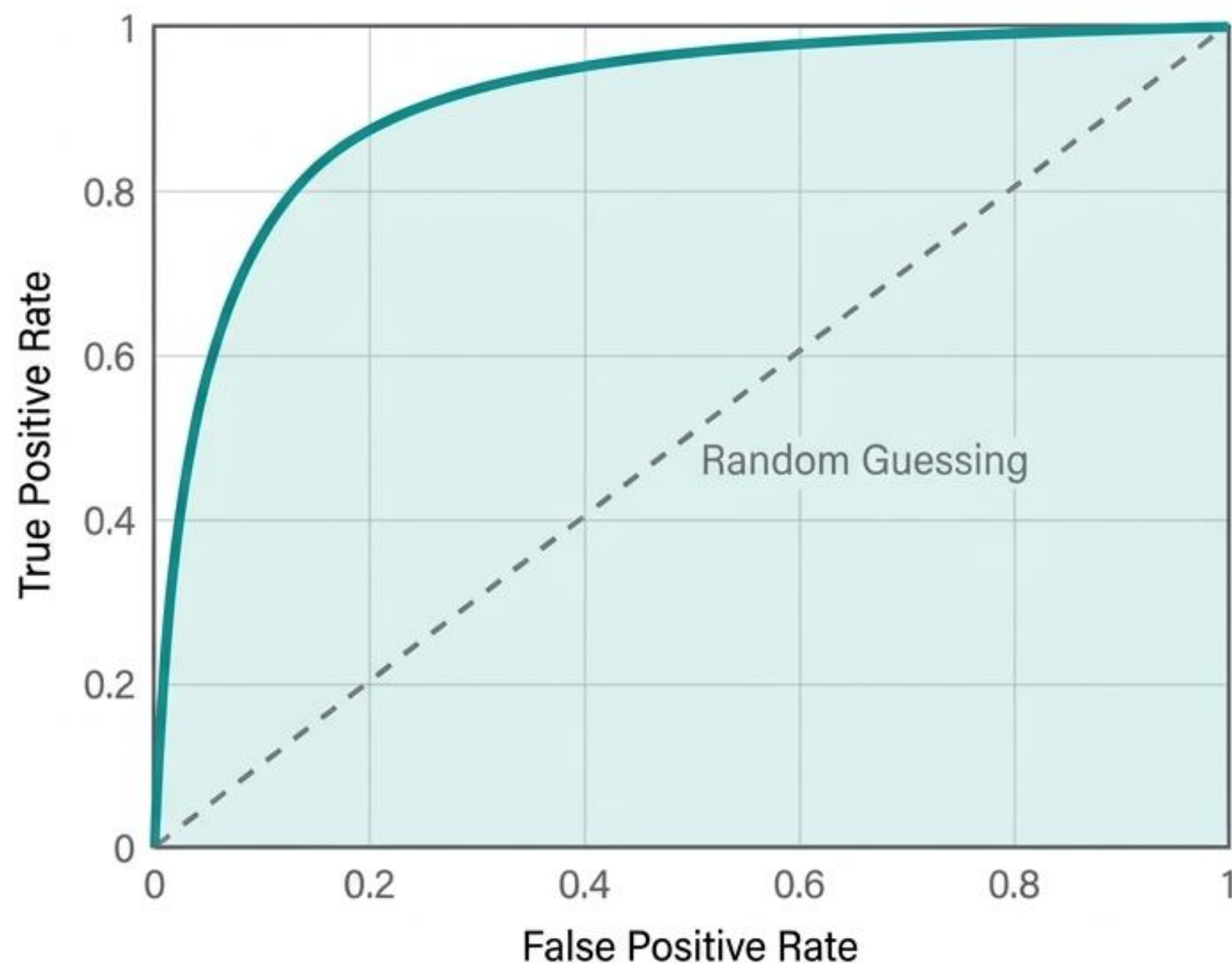
Biomedical Insight: Different mistakes have **drastically different costs**. Missing a disease (False Negative) is often fatal, whereas a false alarm (False Positive) requires a follow-up test.

Prioritizing the Right Metric

Accuracy	90% [(36+54) / 100]	Correct overall?	Useful only when classes and error costs are reasonably balanced.
Precision	85.7% [36 / (36+6)]	When we predict positive, how often are we right?	Important when false positives (false alarms) are highly costly.
Recall / Sensitivity	90% [36 / (36+4)]	How many actual positives did we detect?	Crucial when missing a positive case is costly (e.g., disease screening).
F1-Score	87.8%	Harmonic mean.	Useful when both precision and recall must be balanced.

Discussion Prompt: For your specific biomedical screening problem, which metric will you prioritize and why?

Measuring Discrimination: ROC and AUC



The Concept

ROC-AUC measures how well the model ranks or separates the two classes across all possible classification thresholds.

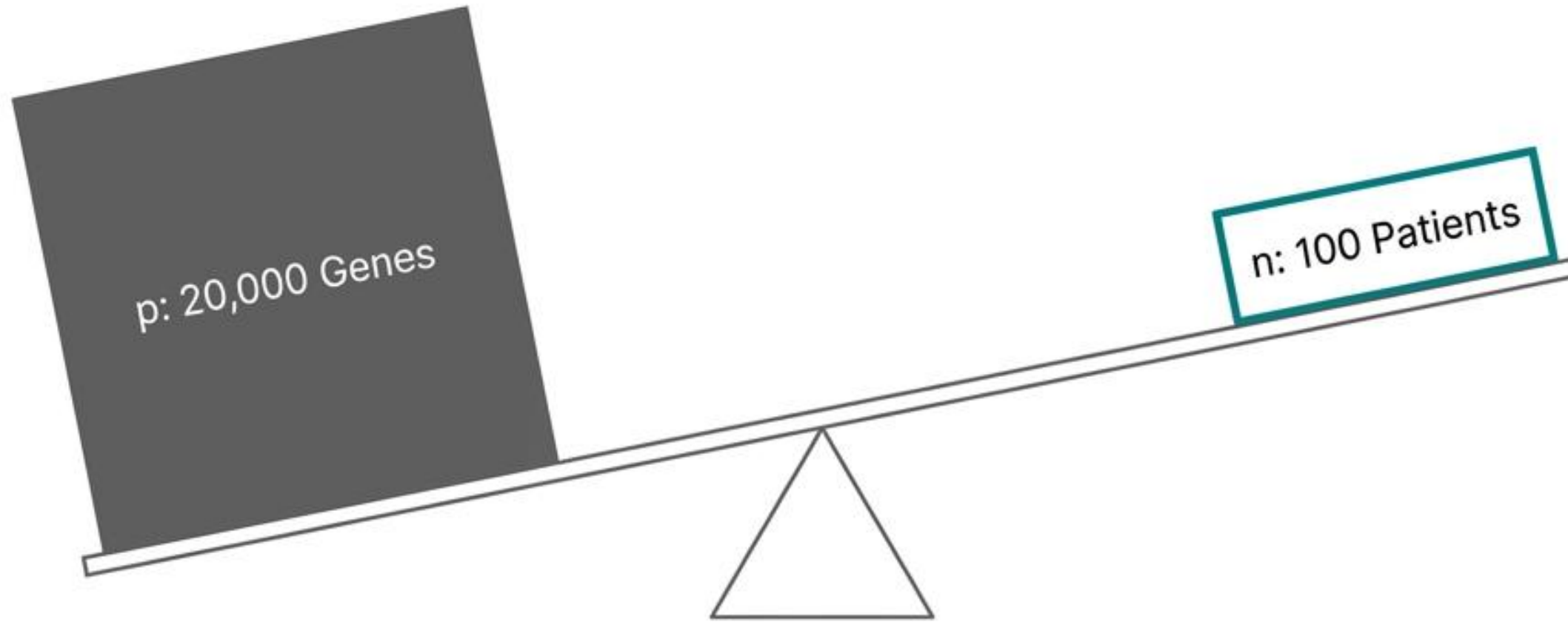
The Goal

Push the curve as close to the top-left corner as possible.

AUC (Area Under the Curve)

Summarizes distinction ability. A curve closer to the top-left indicates superior discrimination. The diagonal represents an AUC of 0.5 (a coin flip).

The Bioinformatics Challenge ($p \gg n$)



The Problem

In omics studies, measured variables (features) vastly exceed the number of patient samples. Data becomes sparse, and models fit noise instead of reproducible biology.

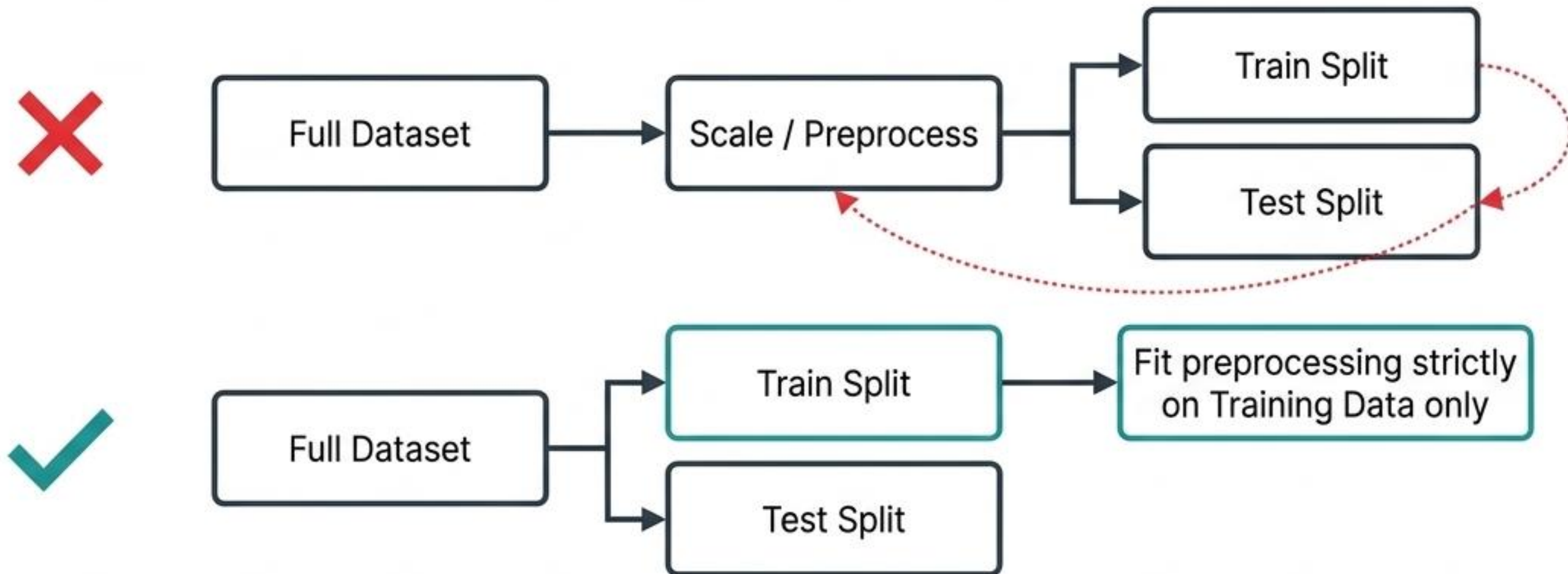
The Risks

Severe overfitting, unstable feature selection, and poor generalization to new cohorts.

The Defenses

- **Feature Selection:** Retain only informative variables.
- **Dimensionality Reduction:** Summarize variables.
- **Regularization:** Penalize overly complex models.

The Danger of Data Leakage



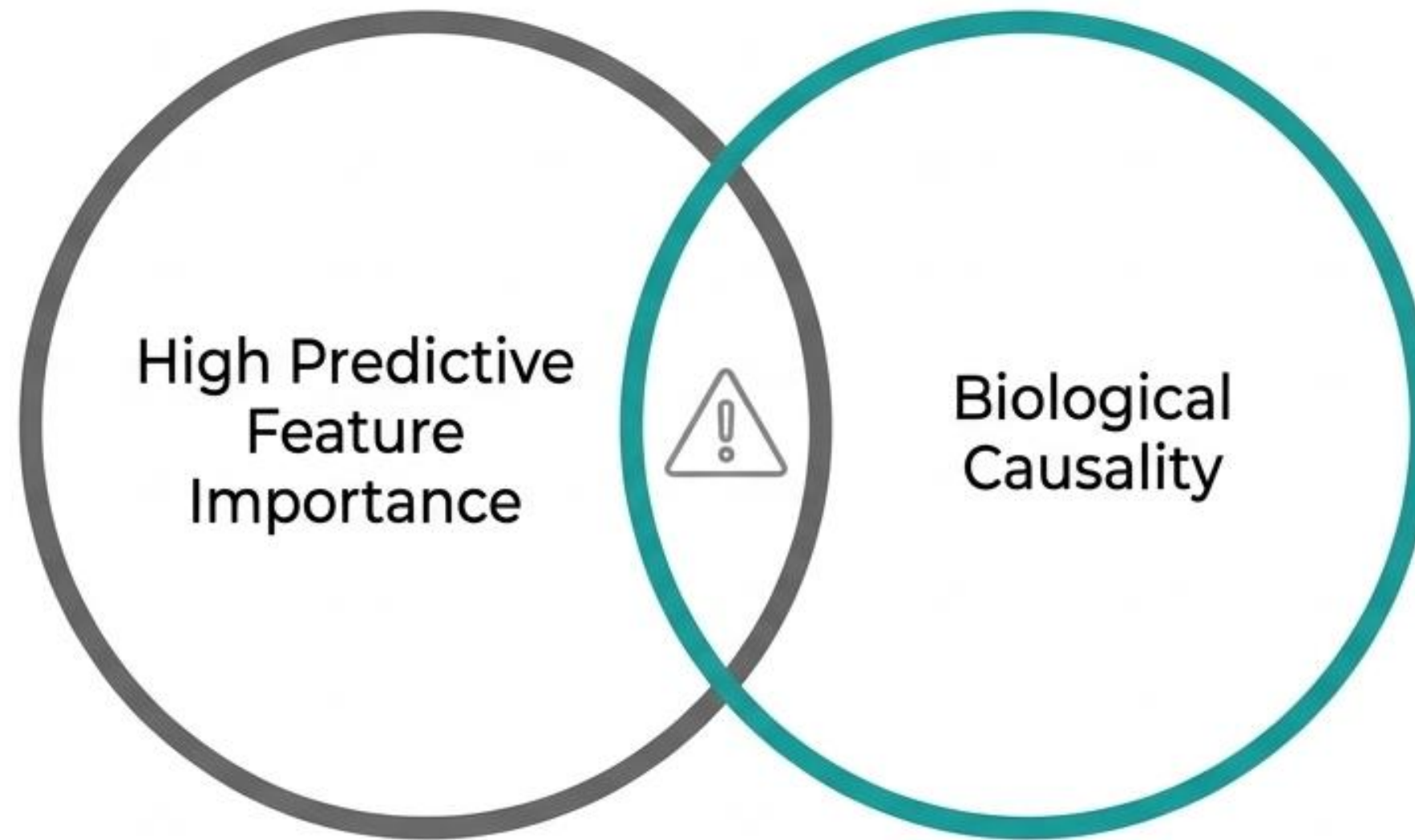
The Error

Leakage occurs when information that should be strictly unavailable during training (like the test set distribution) inadvertently influences the model.

The Rule

Feature selection and all learned preprocessing (like scaling) must happen inside each training fold. Doing this before splitting gives dangerously optimistic results.

Interpretability != Causality



The Concept: A specific feature (e.g., a gene) may be highly useful for making a prediction without being the actual biological cause of the outcome.

The Takeaway: Machine-learning feature importance simply describes a predictive contribution within mathematical parameters. It does not replace lab experiments as causal evidence.

Synthesis: Pre-Lab Checklist

☐

1. Variables Defined: Have you explicitly identified your Features (X) and your Target (y)?

☐

2. Approach Selected: Does your problem require Supervised or Unsupervised learning?

☐

3. Data Secured: Is your Test Data strictly held out to prevent leakage?

☐

4. Dimensionality Managed: Are you prepared for the risks of $p \gg n$ overfitting?

☐

5. Success Defined: Have you chosen the classification metric that reflects true biomedical consequences?

